Fiscal Year:	FY 2026	**Task Last Updated:**	FY 04/10/2026
PI Name:	Karma, Alain Ph.D.		
Project Title:	Computational Modeling of Columnar-Equiaxed Alloy Solidification Micro-Structures (COMPASS)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02115-5005	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	01/15/2024	**End Date:**	01/14/2027
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC24K0466		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The primary objective of the Computational Modeling of Columnar-Equiaxed Alloy Solidification Micro-Structures (COMPASS) project is to develop and validate computational models that accurately predict microstructure evolution during alloy solidification. By employing phasefield and dendritic needle network (DNN) models, the research focuses on key solidification phenomena, including columnar and equiaxed growth, the columnar-to-equiaxed transition (CET), and dendritic spacing selection. A major emphasis is placed on benchmarking these models against microgravity experiments to ensure predictive accuracy under diffusive conditions. Additionally, comparisons between the closed-source graphics processing unit (GPU)-based phase-field solidification research code (GPUPF) and the open-source PRISMS-PF framework further strengthen the computational foundation for alloy solidification modeling.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	The major accomplishments during the present grant period include: • A critical correction was made to the thermophysical parameters used in DNN simulations of the NPG-20wt%DC alloy system. • Building on the corrected material parameters, we successfully implemented and executed DNN simulations with time-varying pulling velocities matching the step-change protocols employed in the CETSOL1 experiments. • Additionally, we quantitatively investigated equiaxed solidification in NPG–20wt.%DC alloys under isothermal and time-dependent thermal conditions relevant to the TRANSPARENT-CETSOL2 campaign using the DNN method. • The phase-field simulations were carried out to support ongoing investigations of dendrite orientation transitions (DOTs) in NPG–DC alloys. • A significant advancement was achieved in the phase-field modeling of the Device for the Study of Critical Liquids and Crystallization (DECLIC) Directional Solidification Insert-R (DSIR) experiments through the development and implementation of a ternary alloy solidification model for the SCN-camphor-water system. • The detailed benchmarking study comparing the GPU-based finite-difference code (GPUPF) with the open-source PRISMS-PF framework was completed and refined during this reporting period.
Bibliography Type:	Description: (Last Updated: 05/10/2026)